

APEXPLANET SOFTWARE PVT. LTD.

DATA ANALYTICS INTERNSHIP - TASK 5

Final Reporting, Deployment & Presentation

Project: Superstore Sales and Profit Analysis

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Internship Period: 11 July 2026 - 24 August 2026

Dataset: cleaned_superstore.csv

Submission-ready report aligned to the official Task 5 activity sequence.

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1. Executive Summary

This Task 5 report converts the supplied cleaned Superstore dataset into an executive-ready analytics summary. The file contains **10 records** across **8 columns**. Verified totals are **4,760** sales, **930** profit and **31** units across **10** orders, with an overall profit margin of **19.54%**.

Technology is the largest category by sales (2,350) and profit (570). East is the largest region by sales (1,775), while South has the highest regional margin (23.70%). North has the weakest regional margin (6.25%). Table is the only loss-making sub-category in the supplied records.

This report also documents the automation pipeline, README/GitHub structure, presentation plan, demo-video script and final submission checklist required by the official Task 5 activity sequence.

2. Problem Statement & Objectives

Problem: Business users need a clear view of revenue and profitability by category, sub-category and region so they can identify strong performers, loss-making areas and opportunities for improvement.

Objectives: (1) produce an executive summary; (2) validate the cleaned dataset; (3) identify 5-7 evidence-based insights; (4) document statistical analysis; (5) include actual Power BI dashboard evidence; (6) prepare an automated analytics pipeline; (7) prepare README/GitHub documentation; (8) provide a presentation and demo script; and (9) finalize submission artifacts.

3. Data Source & Methodology

Data source: the supplied `cleaned_superstore.csv`. The workflow is: **CSV input -> validation -> cleaning checks -> KPI calculation -> category/region/sub-category aggregation -> statistical analysis -> visualization -> Power BI dashboard -> recommendations -> reporting**.

Tools used for this deliverable include Python/Pandas for validation and calculations, Matplotlib for supporting charts, and Power BI for the supplied dashboard evidence. Git/GitHub and scheduled execution are documented as the deployment/version-control layer.

4. Data Quality & Preprocessing

Check	Verified result
Rows / columns	10 / 8
Missing cells	0
Duplicate rows	0
Unique orders	10
Date parsing	Successful
Negative-profit rows	1

The negative-profit record is retained because it reflects business performance. No missing-cell or duplicate-row cleanup was required in the supplied file.

5. Key Findings with Visuals

Finding 1: Technology leads category sales and profit. Sales = 2,350; Profit = 570.

Finding 2: Furniture is a major revenue contributor but has a lower margin (14.67%) than Technology.

Finding 3: East is the largest region by sales (1,775).

Finding 4: South has the highest regional profit margin (23.70%).

Finding 5: North has the weakest regional margin (6.25%).

Finding 6: Table is the only loss-making sub-category in the supplied records (profit -20).

Finding 7: Office Supplies has 19 units but only 160 sales, indicating a low-value/high-volume pattern in this small sample.

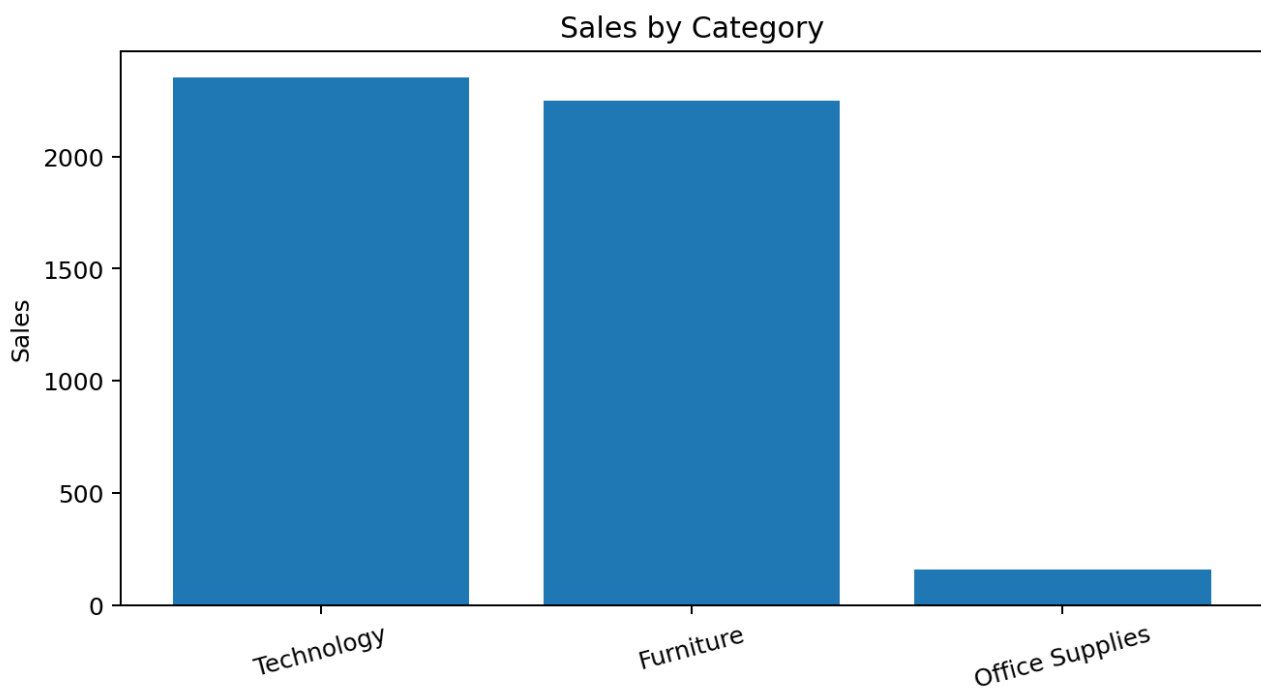


Figure 1. Sales by Category

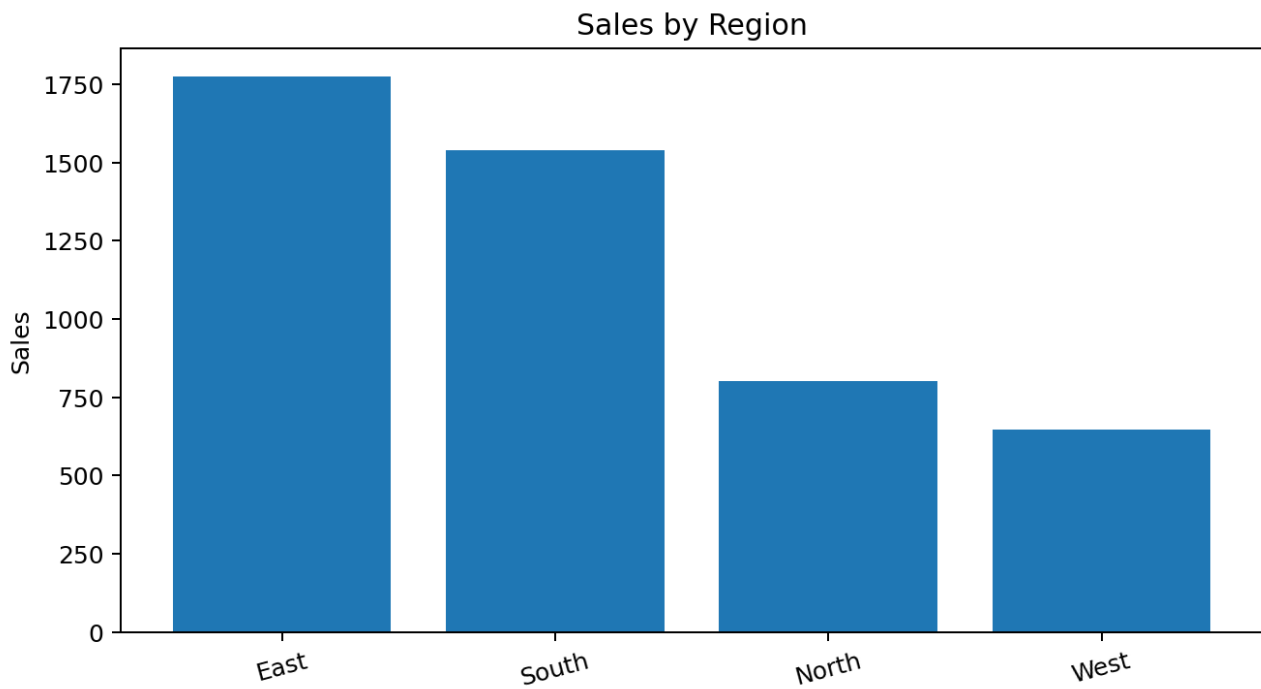


Figure 2. Sales by Region

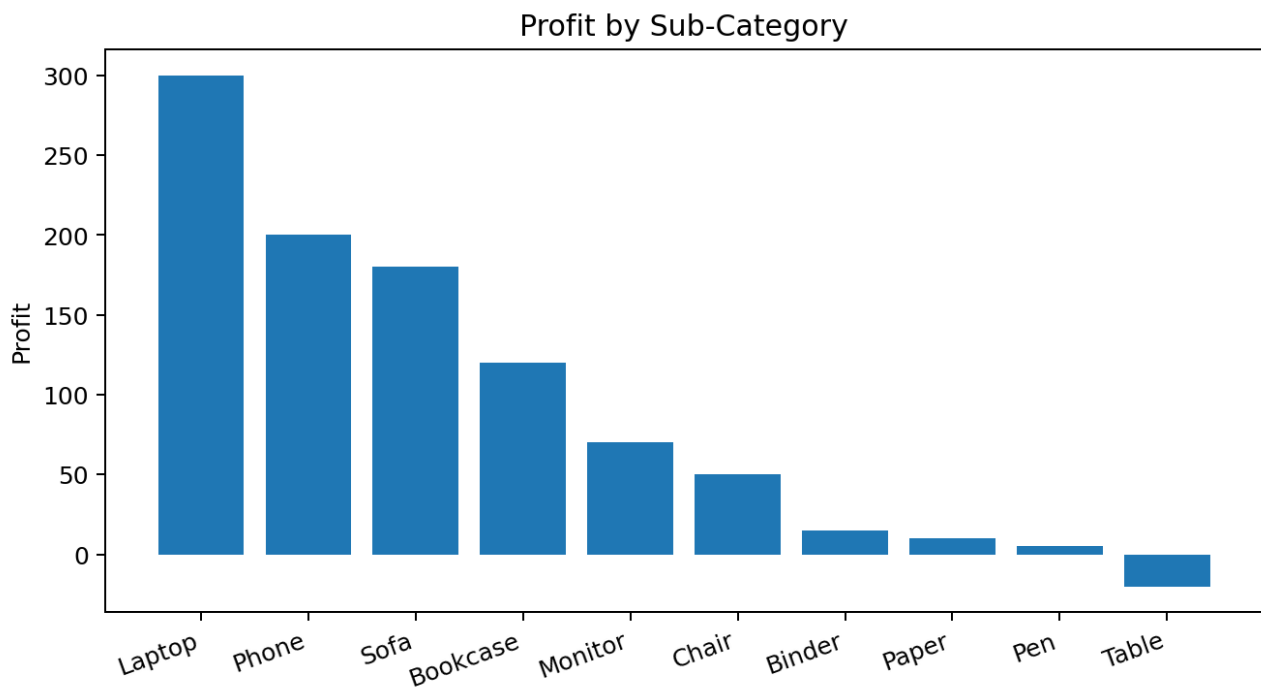


Figure 3. Profit by Sub-Category

6. Statistical Analysis

Statistic	Sales	Profit	Quantity
Mean	476.00	93.00	3.10
Median	400.00	60.00	2.00
Std. deviation	404.83	104.59	2.77
Minimum	25.00	-20.00	1.00
Maximum	1200.00	300.00	10.00

Sales-profit correlation = **0.919**. Quantity-sales correlation = **-0.587**. Because the supplied dataset is only 10 records, correlation results are descriptive rather than predictive or causal.

The official Task 5 reference also mentions time-series forecasting, customer segmentation and churn prediction. Those outputs are intentionally not stated as completed here because the corresponding model files/results were not supplied in the current evidence.

7. Dashboard Evidence

The screenshot below is the actual Power BI dashboard supplied in this conversation. It is shown with the category filter cleared, so all three categories are visible.

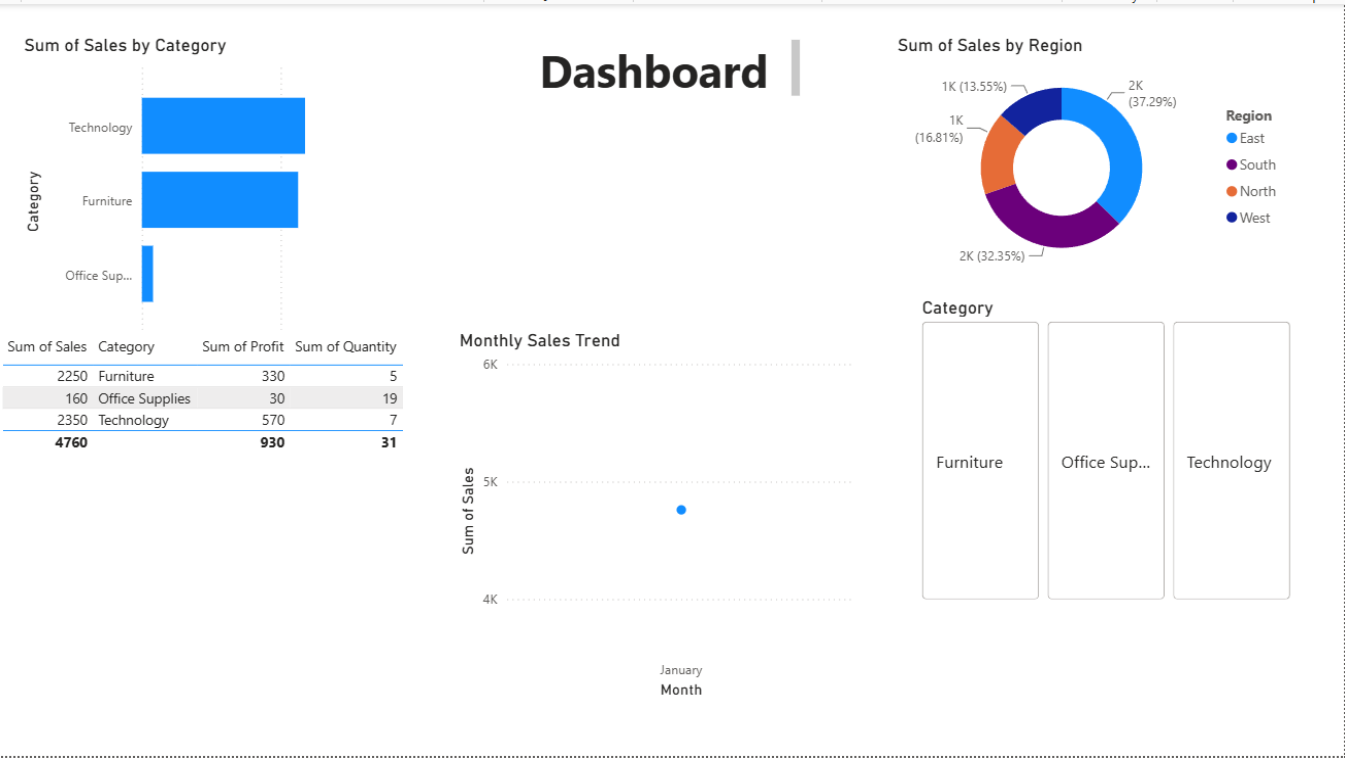


Figure 4. Actual Power BI Dashboard

Dashboard KPI values: Total Sales = 4,760, Total Profit = 930, Total Quantity = 31. Category totals shown in the dashboard are Furniture 2,250, Office Supplies 160, and Technology 2,350.

8. Business Recommendations

Recommendation 1: Prioritize Technology products because they lead both sales and profit in the supplied data.

Recommendation 2: Review Furniture pricing, procurement and fulfillment economics to protect margin.

Recommendation 3: Investigate North region performance before increasing sales investment; compare product mix and costs with stronger regions.

Recommendation 4: Review the Table sub-category for pricing, purchase cost and fulfillment issues because the supplied Table record is loss-making.

Recommendation 5: Improve Office Supplies basket value through bundling/cross-selling, while monitoring whether higher unit volume converts into revenue.

Recommendation 6: Use the Power BI dashboard as the recurring executive monitoring layer for sales, profit and category/region performance.

Recommendation 7: Use the automation script for repeatable KPI refreshes rather than manually rebuilding summaries.

9. Automated Analytics Pipeline

Day 41-42 activity is addressed with the prepared **task5_automation.py**. The script reads the dataset, validates required columns, checks data quality, calculates KPIs, creates category/region/sub-category summaries and exports results to CSV files.

Scheduling: the script is ready to be scheduled with Windows Task Scheduler or adapted to GitHub Actions/Airflow. A scheduler registration is not claimed as completed in this report.

Outputs: kpi_summary.csv, category_summary.csv, region_summary.csv, subcategory_summary.csv and run_log.txt.

10. GitHub Documentation

A clean repository structure has been prepared with the automation script, requirements file, README, demo script, report and presentation. The final GitHub push must be completed from the user's GitHub account.

Suggested structure:

data/cleaned_superstore.csv

scripts/task5_automation.py

outputs/

requirements.txt

README.md

demo_video_script.txt

ApexPlanet_Task5_Final_Report_Corrected.pdf

ApexPlanet_Task5_Presentation_Corrected.pptx

11. Demo Video & Presentation

The official Task 5 asks for a 5-7 minute demo video and an 8-10 slide presentation. A ready-to-use demo script is included with this package.

Demo flow: Introduction -> Problem statement -> Data cleaning -> EDA insights -> Power BI dashboard -> Statistical analysis -> Automation pipeline -> Recommendations -> Conclusion.

Presentation plan: 1) Title, 2) Problem & Objectives, 3) Data Overview, 4) EDA Insights, 5) Dashboard Analysis, 6) Statistical Results, 7) Business Recommendations, 8) Challenges & Learnings, 9) Future Scope, 10) Thank You.

Actual recording and upload are manual final steps and are not falsely marked as completed.

12. Final Submission Checklist

- Final report PDF - prepared.
- Power BI dashboard evidence - included.
- Automation script - prepared.
- requirements.txt - prepared.
- README.md - prepared.
- Presentation deck - prepared.
- Demo video script - prepared.
- GitHub repository push - manual step remaining.
- LinkedIn post / link - manual step if required.
- Demo video recording and upload - manual step remaining.
- Dashboard public link - manual step if the organization requires it.

13. Future Scope

With a longer historical dataset, the project can be extended to sales forecasting, customer segmentation, churn prediction, anomaly detection and automated alerts. Cloud deployment can also be introduced for scheduled refreshes and shared dashboards.

14. Limitations

The supplied dataset contains only 10 records covering a short date range. It also does not include several fields mentioned by the internship reference, such as customer segmentation/churn inputs or model outputs. Therefore this report avoids unsupported claims and treats the statistical findings as descriptive.

15. Conclusion

Task 5 has been structured around the actual requirements: executive reporting, key findings, statistical analysis, real dashboard evidence, automation documentation, GitHub/README preparation, demo/presentation planning and final submission readiness. The verified dataset totals are 4,760 sales, 930 profit and 31 units. The strongest immediate opportunities are Technology prioritization, North-region review and loss-making Table analysis.

16. Appendix

Core category table

Category	Sales	Profit	Qty	Margin
Technology	2,350	570	7	24.26%
Furniture	2,250	330	5	14.67%
Office Supplies	160	30	19	18.75%